

Extending RF Fingerprinting-Based Physical Layer Authentication Through Additional Machine-Learning Models

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Abstract

This project studies RF fingerprinting for Physical Layer Authentication (PLA) using the reference implementation associated with the paper by Alla et al. [1]. The reference work investigates whether hardware impairments, Gabor-transform based signal processing, and diagonal spectrum reading can produce RF fingerprints that are useful for device recognition. The purpose of this project is not to replace the full PLA framework, but to keep its assessment logic and extend the machine-learning comparison with two additional classifiers: Gaussian Naive Bayes and Extra Trees. The experiments use the processed RF fingerprint dataset provided in the public PLA repository [2]. Results show that the added models can reach high Average Detection Rate (ADR) values, but Random Forest remains the strongest baseline in the supplied results.

1 Introduction

Wireless devices contain small hardware imperfections introduced by their radio front-end components. These imperfections can slightly affect the transmitted signal and can therefore be used as device-specific physical-layer characteristics. RF fingerprinting uses this idea to recognize devices from their signal behavior rather than only from higher-layer identifiers.

The reference paper by Alla et al. [1] studies a machine-learning assisted PLA framework based on three hardware impairments: Carrier Frequency Offset (CFO), Direct Current Offset (DCO), and Phase Offset (PO). The paper combines these impairments with Gabor-transform based processing and diagonal reading of the spectrum to generate RF fingerprints. The machine-learning stage then evaluates which classifiers and selected features are more robust for RF fingerprinting recognition.

This project follows the same assessment direction. The main contribution is the addition of two extra machine-learning models to the comparison, while keeping the same processed RF fingerprint dataset, the same trial structure, and the same evaluation criteria used by the reference code.

2 Reference PLA Methodology

The reference PLA workflow can be summarized as follows:

1. Signals are represented as I/Q samples related to wireless transmissions.
2. Hardware impairments are considered: CFO, DCO, and PO.
3. Gabor-transform based processing is used to represent signal behavior in the time-frequency domain.
4. Diagonal reading of the spectrum is used to extract fingerprint-relevant information.
5. Statistical features are generated from the resulting representation.
6. Feature-selection methods are applied before model training.
7. Machine-learning classifiers are assessed using authorized and rogue-device scenarios.

The objective is an assessment of ML models and feature-selection approaches for RF fingerprinting recognition. At this stage, the purpose is not to assume in advance a final real-world deployment model, but to evaluate whether the fingerprints and classifiers can satisfy the target criteria used in the reference code.

3 Dataset

The project uses the processed dataset included in the PLA repository. The dataset contains 146 processed RF fingerprint samples, 505 features per sample, and 12 device labels. The larger raw signal dataset and noise dataset are not included in this submission because of their size.

Table 1: Dataset summary

Item	Value
Processed samples	146
Features per sample	505
Devices	12
Device labels	device1–device12

Table 2: Samples per device in the processed dataset

Device	Label ID	Samples
device1	0	13
device2	1	14
device3	2	13
device4	3	15
device5	4	13
device6	5	17
device7	6	13
device8	7	15
device9	8	13
device10	9	8
device11	10	6
device12	11	6

4 Trial Structure

The same three trial scenarios from the PLA code are used. The trials change the number of authorized and rogue devices, allowing the ML models to be assessed under different fingerprinting conditions.

Table 3: Trial structure used in the PLA assessment

Trial	Authorized devices	Rogue devices
trial_1	device3, device2, device12, device9	device8, device11, device5, device1, device6, device10, device7, device4
trial_2	device10, device6, device12, device3, device11, device1	device2, device5, device8, device7, device4, device9
trial_3	device1, device10, device9, device8, device12, device11, device6, device7	device4, device2, device5, device3

For each authorized device, the classifier is trained to distinguish the target authorized device from the other authorized devices in the same trial. Rogue devices are then evaluated according to the same criteria used in the reference code. The target values are TDR at least 95% and FDR at most 5%.

5 Additional Machine-Learning Models

Two additional models were selected for this project.

5.1 Gaussian Naive Bayes

Gaussian Naive Bayes is a probabilistic classifier based on Bayes’ theorem with a Gaussian assumption for feature distributions. It was selected because it is simple, fast, and provides a useful lightweight baseline. Since the processed RF fingerprint dataset has a limited number of samples compared with the number of features, a simple probabilistic model is useful for checking whether the feature representation is already discriminative enough without requiring a complex model.

5.2 Extra Trees Classifier

Extra Trees is an ensemble method related to Random Forest, but it introduces stronger randomization when building decision trees. It was selected because the strongest supplied baseline is Random Forest, so Extra Trees is a natural comparison. The goal is to test whether a more randomized tree ensemble can offer similar or better robustness for the same RF fingerprinting assessment.

6 Feature-Selection Setting

The feature count is not treated as a fixed assumption. Instead, the added models are evaluated over a feature-count grid:

1, 5, 10, 25, 50, 75, 100, 150, 200, 300, 400, 505.

The best value is selected according to the same objective used in the reference code: satisfying the TDR/FDR criteria when possible, otherwise choosing the configuration closest to the target thresholds. This keeps the additional-model evaluation consistent with the reference logic and prevents the comparison from depending on a single arbitrary feature-count value.

The executed additional-model experiment uses ANOVA feature selection. ANOVA was selected because it is the strongest feature-selection setting among the supplied original results, where Random Forest with ANOVA reaches the best mean ADR.

7 Results

7.1 Original Repository Results

The supplied repository results show that Random Forest with ANOVA is the strongest baseline. Table 4 summarizes the best original combinations.

Table 4: Selected original repository results, averaged across the three trials

Model	Feature method	Mean ADR
Random Forest	ANOVA	98.38%
Random Forest	Mutual Information	97.55%
Logistic Regression	ANOVA	97.21%
Logistic Regression	PCA	96.32%
Gradient Boosting	Mutual Information	95.40%
Random Forest	RFE	95.23%
Gradient Boosting	ANOVA	94.48%
XGBoost	Mutual Information	94.46%

7.2 Added Models

Table 5 shows the two added models using the feature-count search with ANOVA.

Table 5: Added-model results with ANOVA feature selection

Model	Mean ADR
Extra Trees	94.90%
Gaussian Naive Bayes	94.82%

Both added models perform competitively, but neither outperforms the strongest original result. Extra Trees is close to several original baselines, which is reasonable because it is related to Random Forest. Gaussian Naive Bayes also reaches a high ADR, showing that the selected RF features contain useful discriminative information even for a simple probabilistic classifier.

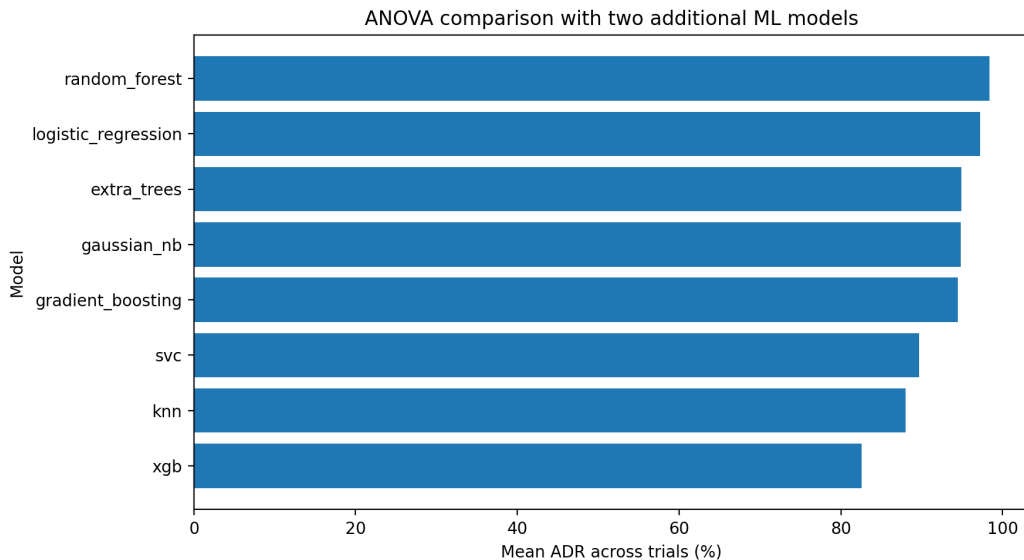


Figure 1: ANOVA comparison between supplied repository models and the two added models.

7.3 Device-Level Results

The added-model evaluation also records the selected number of features per target device. This gives a more detailed view of how the models behave across the trial scenarios.

Table 6: Example device-level results for trial 2

Model	Target device	Features	Auth. TDR	Rogue TDR	ADR
Gaussian NB	device10	1	100.00%	96.39%	98.19%
Gaussian NB	device6	1	100.00%	85.54%	92.77%
Gaussian NB	device12	50	100.00%	96.39%	98.19%
Gaussian NB	device3	5	100.00%	67.47%	83.73%
Extra Trees	device10	5	100.00%	93.98%	96.99%
Extra Trees	device6	50	100.00%	91.57%	95.78%
Extra Trees	device12	25	100.00%	100.00%	100.00%
Extra Trees	device3	100	100.00%	97.59%	98.80%

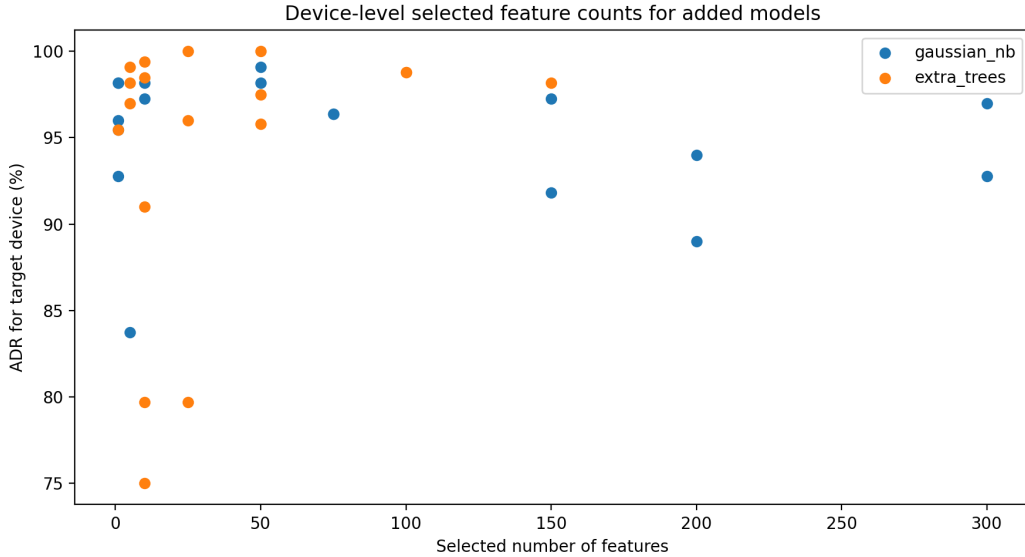


Figure 2: Selected feature counts and device-level ADR for the two added models.

8 Discussion

The results support three observations. First, the original Random Forest baseline remains the best-performing model in the supplied result set. Second, Extra Trees is a suitable comparison model because it belongs to the same family of tree ensembles and achieves a competitive mean ADR. Third, Gaussian Naive Bayes performs surprisingly well for such a simple classifier, suggesting that the RF fingerprint features already contain relevant separation between devices.

The project does not claim that the added models improve the state of the art. Instead, it presents them as additional comparative baselines inside the same PLA assessment framework. This is consistent with the objective of assessing models and feature selections rather than proposing a final deployed malicious-device detector.

9 Limitations

The main limitation is that the additional-model execution focuses on ANOVA feature selection. This was chosen because the supplied original results show ANOVA as the strongest feature-selection method for the best baseline model. A fuller continuation would run the added models with Mutual Information, PCA, and RFE as well, using the same feature-count search logic.

The second limitation is that the submitted repository uses the processed RF fingerprint dataset only. The larger raw I/Q dataset and the noise dataset are not included because of their size. Therefore, the report discusses the supplied noise summaries but does not claim new SNR experiments for the two added models.

10 Conclusion

This project follows the objective of the reference paper by evaluating RF fingerprinting recognition through hardware impairments, Gabor-transform based processing, diagonal spectrum reading, feature selection, and ML model assessment. It cites the reference work as a joint research paper and evaluates two additional ML models inside the same trial structure.

The main result is that Extra Trees and Gaussian Naive Bayes reach competitive ADR values of 94.90% and 94.82%, respectively, with ANOVA feature selection. However, Random Forest with ANOVA remains the strongest supplied baseline at 98.38%. The result is still useful because it shows that not all additional ML models improve RF fingerprinting performance, and that model suitability must be evaluated empirically for this type of physical-layer authentication task.

References

- [1] I. Alla, S. Yahia, V. Loscri, and H. Eldeeb. *Robust Device Authentication in Multi-Node Networks: ML-Assisted Hybrid PLA Exploiting Hardware Impairments*. Proceedings of the 2024 Annual Computer Security Applications Conference, pp. 1172–1185, 2024.
- [2] PLA-AP. *Physical Layer Authentication Repository*. GitHub repository: <https://github.com/PLA-AP/PLA>.